Younous Maillefer

St-Sulpice (VD), Switzerland • younous.maillefer@ik.me • github.com/Homotopic-to-the-circle/Academia

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland
Master of Science in Mathematics *Sept 2023 – Feb 2026*

- *Pure mathematics:* Riemann Surfaces, Topology, Representation Theory, Graph Theory, Abstract Analysis on Groups, Number Theory.
- *Applied mathematics:* Number Theory in Cryptography, Probability Methods in Modern AI, Optimization on Manifolds, Lattice Models, Error Control in Scientific Modelling.

École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland
Bachelor of Science in Physics *Sept 2019 – July 2023*

- *Mathematics:* Commutative Algebra, Lie Algebras, Functional Analysis, Differential Geometry, Homology, Metric Spaces.
- *Physics:* Quantum Physics, Classical Electrodynamics, Analytical Mechanics, Statistical Physics, Solid-State Physics, Particle Physics, Computational Physics, Philosophy of the Exact Sciences.

RESEARCH EXPERIENCE

Master Valorisation (post-thesis research) EPFL
Section of Mathematics *Mar 2026 – Aug 2026*

- Extended the thesis results on cyclotomic shortness with a view to publication (manuscript in preparation).
- Independent study of artificial life.

Master's Thesis: *Bounds and Structure of the Cyclotomic Shortness of* Φ_1^N EPFL
Supervisor: Prof. Maryna Viazovska • Expert: Prof. James McKee • Grade: 6.0/6.0 *Autumn 2025*

- Studied the minimal ℓ^1-norm of cyclotomic multiples of $(z-1)^N$, motivated by compact representations of polynomials of small Mahler measure.
- Derived a strict exponential upper bound using Euler's Pentagonal Number Theorem and Sudler's constant.
- Proved, by evaluation at roots of unity, that certain cyclotomic factors are forced in every short polynomial, and used the uncertainty principle on finite abelian groups to force further structure.

Semester Project: *DFTK Refinement – Verification Across a Large Set of Systems* EPFL
Supervisor: Prof. Michael Herbst • Co-supervisor: Bruno Ploumhans • Grade: 5.5/6.0 *Spring 2025*

- Implemented and validated error estimates in DFTK.jl, the Julia density-functional toolkit.
- Automated equation-of-state benchmarks over a large set of systems on the EPFL HPC cluster.
- Quantified the trade-off between kinetic-energy cutoff and numerical accuracy, and derived guidelines for choosing computational parameters.

TEACHING EXPERIENCE

EPFL Lausanne, Switzerland
Teaching Assistant *2021 – 2026*

- Analysis I & II, Linear Algebra I & II, Classical Mechanics.

Gymnasiums of Biel, Payerne and Neuchâtel Switzerland
Substitute Teacher, Mathematics and Physics *2018 – Present*

- Taught upper-secondary classes; prepared lectures, exercise sets and exams.

SKILLS AND LANGUAGES

Programming	Julia (advanced), Python (NumPy, PyTorch), C++, MATLAB, SageMath, Lean (basic)
Tools	Machine learning (neural networks), HPC clusters (Linux, Bash), Git, LaTeX
Languages	French (native), English (C1), German (B2+)